

CSC4026Z Chat Server Documentation

Channel Messages

These messages deal with Channels. The definitions below use [these type indicators](#).

CHANNEL_CREATE

Use this to create a new channel. When you create a new channel, you are also joined to it. You **MAY** set a channel description, which is visible through the [CHANNEL_INFO](#) message. The server will return an error if you try to create more than **20** channels. The server will handle up to **1000 channels across all users**.

Request

```
CHANNEL_CREATE_request {
    u8      request_type = 4
    u32     session
    u32     request_handle
    s[20]   channel      # Name of new channel
    s[100]  description  # A description associated with the
}
```

The `name` can only contain letters, numbers, underscores, and dashes, and is limited to 20 characters.

Response

```
CHANNEL_CREATE_response {
    u8      response_type = 25
    u32     session
    u32     response_handle
    s[20]   channel      # Name of new channel
    s[100]  description  # A description associated with the
}
```

CHANNEL_LIST

Use this to fetch a list of all current channels. A full channel list won't always fit in a single UDP packet, so the message allows for rudimentary pagination. The server will return a list of up to 10 channel names. If there are more than 10 channels on the server, then the `next_page` field will be `True`. You **MAY** repeat the list request, and include the `offset = 10`, to skip the first

10 channels, `offset = 20` to skip the first 20 channels, and so on.

Note that the server's channel list **MAY** change between your list requests; if a channel was removed between your *CHANNEL_LIST* requests, then your view of the list will not be accurate. Such is life.

Request

```
CHANNEL_LIST_request {
    u8      request_type = 5
    u32     session
    u32     request_handle
    u16     offset          # Leave out unless you are paging through t
}
```

Response

```
CHANNEL_LIST_response {
    u8      response_type = 26
    u32     session
    u32     response_handle
    [s[20]] channels        # List of channels
    bool    next_page       # Flag indicating whether more chan
}
```

CHANNEL_INFO

Use this to query information about a Channel.

Request

```
CHANNEL_INFO_request {
    u8      request_type = 6
    u32     session
    u32     request_handle
    s[20]   channel        # Name of channel to query
}
```

Response

```
CHANNEL_INFO_response {
    u8      response_type = 27
    u32     session
    u32     response_handle
    s[20]   channel        # Channel the user is querying
    s[100]  description    # A description associated with the
    [s[20]] members        # List of channel members
}
```

CHANNEL_JOIN

Use this to join a Channel.

Request

```
CHANNEL_JOIN_request {
    u8      request_type = 7
    u32      session
    u32      request_handle
    s[20]    channel      # Name of channel to join
}
```

Response

This message **MAY** arrive unsolicited. If you are joined to a channel where another user joins, then you will receive this message to inform you a new user has arrived. In those cases, `response_handle` will be `None`.

```
CHANNEL_JOIN_response {
    u8      response_type = 28
    u32      session
    u32      response_handle  # Only filled in for the sender
    s[20]    username        # Name of joining user
    s[20]    channel         # Channel the user is joining
    s[100]   description     # A description associated with the
}
```

CHANNEL_LEAVE

Use this to exit a Channel. If you are the last user in the channel, then the server will remove the channel as you exit.

Request

```
CHANNEL_LEFT_request {
    u8      request_type = 8
    u32      session
    u32      request_handle
    s[20]    channel      # Name of channel to Exit
}
```

Response

This message **MAY** arrive unsolicited. If you are joined to a channel where another user leaves, then you will receive this message to inform you they are no longer in the channel. In those cases, `response_handle` will be `None`.

```
CHANNEL_LEFT_response {
    u8      response_type = 29
    u32      session
    u32      response_handle  # Only filled in for the sender
}
```

```
s[20]    username    # Name of leaving user
s[20]    channel     # Channel the user is leaving
}
```

CHANNEL_MESSAGE

Use this to send a message to a channel. All users who are in the channel will receive the message.

Messages are UTF-8 encoded.

Request

```
CHANNEL_MESSAGE_request {
  u8      request_type = 9
  u32     session
  u32     request_handle
  s[20]   channel      # Name of channel to send message to
  s[500]  message      # Message to send to channel
}
```

Response

This message **MAY** arrive unsolicited. If you are joined to a channel where other users are chatting, you will receive this message frequently. In those cases, `response_handle` will be `None`.

```
CHANNEL_MESSAGE_response {
  u8      response_type = 30
  u32     session
  u32     response_handle # Only filled in for the sender
  s[20]   username      # Name of sending user
  s[20]   channel       # Channel the message was sent to
  s[500]  message       # Message sent to the Channel
}
```

Move on to the [User messages](#).